

A Counterexample to Conjecture 00000003967: the 4-Cycle Attains the Pebbling Bound 2^d

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Abstract

Conjecture 00000003967 claims that the graphs whose pebbling number equals 2^d , with d the diameter, are exactly the paths, and that every 2-connected graph satisfies $\pi(G) \leq 2^d(1 - c/\log d)$ for a constant $c > 0$. The 4-cycle C_4 has diameter 2 and $\pi(C_4) = 4 = 2^2$. It is not a path and it is 2-connected, so both claims fail. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000003967 reads:

Definition: The pebbling number $\pi(G)$ is the minimal number of pebbles guaranteeing that a pebble can be moved to any target vertex. Conjecture: The graphs attaining the Graham bound 2^d are exactly the $(d - 1)$ -subdivision classes of paths, and every 2-connected graph satisfies $\pi(G) \leq 2^d(1 - c/\log d)$.

The Chinese version in `conjectures/00000003967.md` states the same two claims.

2 Reading of the statement

Definition 1. Let G be a finite graph. A *distribution* p assigns a number of pebbles $p(v) \geq 0$ to each vertex v . A *pebbling move* along an edge uv with $p(u) \geq 2$ removes two pebbles from u and adds one pebble to v . A distribution is *r -solvable* if a sequence of moves leads to a distribution with at least one pebble on the target r . The *pebbling number* $\pi(G)$ is the least t such that every distribution of t pebbles is r -solvable for every target r .

Here $d = \text{diam}(G)$ is the diameter. The “Graham bound” 2^d is the classical bound $\pi(G) \geq 2^{\text{diam}(G)}$: put $2^d - 1$ pebbles on a vertex at distance d from the target. A graph *attains* it if $\pi(G) = 2^{\text{diam}(G)}$. Subdividing edges of a path gives again a path, so the class in the first claim consists of the paths. A graph is *2-connected* if it has at least three vertices and stays connected after deleting any one vertex. The two claims are:

- (C1) a connected graph satisfies $\pi(G) = 2^{\text{diam}(G)}$ if and only if it is a path;
- (C2) for some constant $c > 0$, every 2-connected graph with $d = \text{diam}(G) \geq 2$ satisfies $\pi(G) \leq 2^d(1 - c/\log d)$.

Requiring $d \geq 2$ in (C2) avoids $\log 1 = 0$ and only weakens the claim.

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3 Result

Let C_4 be the cycle $0 - 1 - 2 - 3 - 0$.

Theorem 2. $\text{diam}(C_4) = 2$ and $\pi(C_4) = 4 = 2^{\text{diam}(C_4)}$. Moreover C_4 is 2-connected and is not a path. Hence (C1) fails, and (C2) fails for every $c > 0$ because $4 > 4(1 - c/\log 2)$. In particular Conjecture 00000003967 is false.

4 Proof

Diameter, connectivity, not a path. Two distinct non-adjacent vertices of C_4 are opposite, at distance 2 through a common neighbour. Deleting a vertex leaves a path on three vertices, which is connected. C_4 has four edges, whereas a path on four vertices has three, so C_4 is not a path.

Lemma 3. Every distribution of four pebbles on C_4 is r -solvable for every target r .

Proof. By symmetry take $r = 0$. Let a, b be the neighbours 1, 3 of 0 and $o = 2$ the opposite vertex, and suppose $p(0) = 0$. If $p(a) \geq 2$ or $p(b) \geq 2$, move a pebble to 0. Otherwise $p(a), p(b) \leq 1$, so $p(o) \geq 2$. If $p(a) = 1$, move one pebble from o to a , which then holds 2, and move on to 0; similarly if $p(b) = 1$. If $p(a) = p(b) = 0$ then $p(o) = 4$; move twice from o to a , then from a to 0. In all cases at most three moves suffice. $\square$

Lemma 4. If $t \leq 3$ pebbles are placed on vertex 2, the target 0 cannot be reached.

Proof. Let $\Psi(p) = 4p(0) + 2p(1) + p(2) + 2p(3)$. A move from u to a neighbour v changes Ψ by $-2w(u) + w(v)$, where $w = (4, 2, 1, 2)$. For every edge of C_4 we have $w(v) \leq 2w(u)$, so Ψ never increases. Initially $\Psi = t \leq 3$, whereas a distribution with a pebble on 0 has $\Psi \geq 4$. $\square$

Proof of Theorem 2. Lemma 3 gives $\pi(C_4) \leq 4$. Lemma 4 shows that for each $t \leq 3$ some distribution of t pebbles is not 0-solvable, so $\pi(C_4) \geq 4$. Thus $\pi(C_4) = 4 = 2^{\text{diam}(C_4)}$ although C_4 is not a path, refuting (C1). Since C_4 is 2-connected with $d = 2$ and $\log 2 > 0$, for every $c > 0$ we have $2^d(1 - c/\log 2) < 4 = \pi(C_4)$, refuting (C2). $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 2 against Mathlib.

- `moveFn`, `Move`, `Solvable` and `pebblingNumber` follow the definition above. Reachability is Mathlib's `Relation.ReflTransGen`, and the pebbling number is `sInf` of the admissible t .
- `TwoConnected`, `Claim1` and `Claim2` are the notions and claims above, for graphs on `Fin n`. Diameter is Mathlib's `SimpleGraph.diam`, paths are Mathlib's `pathGraph`, and isomorphism is `SimpleGraph.Iso`.
- `C4` is Mathlib's `cycleGraph 4`. `diam_C4`, `twoConnected_C4` and `C4_not_path` are the facts in the first paragraph of the proof.
- Lemma 3 is `four_mem`: a depth-bounded search `solve` is proved sound (`solve_sound`) and checked by `decide` on all distributions of four pebbles and all targets (`solve_all`).
- Lemma 4 is `not_solvable`, via `weight_move` and `weight_reach`. Together they give `pebblingNumber_C4`.

- `conjecture_00000003967_false` states $\neg\text{Claim1} \wedge \neg\text{Claim2}$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000003967_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 5. The second claim fails for every diameter, not only for $d = 2$. The weight argument of Lemma 4, with weights $2^{d-\text{dist}(v,r)}$, shows $\pi(G) \geq 2^{\text{diam}(G)}$ for every connected graph. So no 2-connected graph with $d \geq 2$ satisfies $\pi(G) < 2^d$; for instance the even cycle C_{2m} is 2-connected with diameter m . The first claim also has many other counterexamples, for instance the hypercubes Q_d , with $\pi(Q_d) = 2^d$ by Chung’s theorem [1]. Only the C_4 case is formalized.

References

- [1] F. R. K. Chung, *Pebbling in hypercubes*, SIAM J. Discrete Math. 2 (1989), 467–472.